

-192.168.100.190)

192.168.100.128/26
255.255.255.192
256-192==64

192.168.100.128 192.168.100.192
192.168.100.129

192.168.100.190
192.168.100.191

=====
==
3. HR 30 (192.168.100.193--
-192.168.100.222)

192.168.100.192/27
255.255.255.224
256-224==32

192.168.100.192 192.168.100.224
192.168.100.193

192.168.100.222
192.168.100.223

=====
4. Admin 12 (192.168.100.225--
-192.168.100.238)

192.168.100.224/28
255-240==16

192.168.100.224 192.168.100.240
192.168.100.225

192.168.100.238
192.168.100.239

=====
5. Finance 3 (192.168.100.241--
-192.168.100.246)

192.168.100.224/29
255.255.255.248
256-248==8

192.168.100.240 192.168.100.248
192.168.100.241

192.168.100.246
192.168.100.247

=====
6. WAN 1 2 (192.168.100.249-----
-192.168.100.250)
192.168.100.248/30

255.255.255.252
256-252==4

192.168.100.248
192.168.100.249

192.168.100.252

192.168.100.250
192.168.100.251

```
=====
7. WAN 2                2                (192.168.100.253-----
-192.168.100.254)
```

192.168.100.252/30

192.168.100.252
192.168.100.253

192.168.100.254
192.168.100.255

R-A Summary Address---> 10.0.0.0/8 R-B

10.0.0.0/16 0 0000001
10.1.0.0/16
10.2.0.0/16
10.3.0.0/16
10.4.0.0/16
10.255.0.0/16 1 1111111

```
=====
=====
CSMA/CD                      Carrier Sense Multiple Access/Collision Detection
wired LAN    half/full duplex
CSMA/CA                      Carrier Sense Multiple Access/Collision Avoidance
WLAN                      half duplex
```

Duplex Modes:

PC-PC	full duplex
PC-Switch	FD
Switch-Switch	FD
Switch-Router	FD
Router-Router	FD
Hub-Hub	HD
Switch-Hub	HD
PC-Hub	HD

Collision Domain & Broadcast Domain:

Collision Domain: A logical network where data packets can collide with each other when sent on shared medium. eg HUB

Broadcast Domain: is a logical network where data sent by one device will be delivered to all other devices over that network. example--Switch, Bridge

HUB Domain	how many collision Domains	1 Collision
	how many B'cast domains	1 B'cast Domain
Switch port=	number of Collision Domain	Number of active
	how many B'cast domains	1 B'cast Domain
Bridge port=	number of Collision Domain	Number of active
	how many B'cast domains	1 B'cast Domain
Router port=	number of Collision Domain	Number of active
	how many B'cast Domains	Number of active
port=	number of B'cast Domain	

=====

Router Internal Components and Router Booting process:

1. ROM POST, Mini IOS, Boot Loader, ROMMON (ROM Monitor---troubleshooting mode)
2. DRAM Dynamic RAM volatile RAM temporary storage
3. NVRAM Non-Volatile RAM like Hard disk permanent storage
(configuration files) startup-config
4. Flash Non-Volatile Memory contains IOS
5. Motherboard (Circuit Board)

Router Booting Process:

1. ROM----POST (detects/check hardware available in a router, whether they are functional or not?)
2. Boot Loader----- look for IOS in Flash if available then it be copied into DRAM
(for IOS) if not then it will look for TFTP server
if no tftp server available then router will enter into ROMMON mode
3. Once IOS is available, it will be loaded into DRAM.
4. Boot Loader will look for Saved configuration in NVRAM, if available----then file will be copied into DRAM
if no file available ---then it will enter into setup mode

Configuration Register Value:	0x2102	Normal mode
mode	0x2142	password recovery

=====

Interfaces of Router:

1. Console Port RJ-45 to access/configure Switches/Routers
2. Aux Port (old router) to connect modem/to access/configure Switches/Routers
3. Ethernet Interface

ethernet (e0)	LAN Interface	
fast ethernet (fa0/0)	10Mbps	RJ-45
gigabit ethernet (gi0/0)	100Mbps	RJ-45
	1Gbps	RJ-45

- | | | |
|---------------------|---|--|
| 4. Serial Interface | s0/0 or s0/0/0 | WAN interface |
| 5. PRI/BRI
45) | Primary Rate Interface/Basic Rate Interface | (old router RJ-
56Kbps
ISDN 128kbps) |

hyper-terminal

Router Prompts/ modes

Router> user execution mode

show, enable, disable, ping

2. Router# Privilege mode

show, enable, disable, ping, delete, copy

3. Router(config)# Global Configuration mode
global settings

4. Router(config-if)# interface mode
Assign ip address, shutdown/no shutdown

5. Router(config-line)# Line mode
telnet, ssh, password

5. Router(config-router)# routing configuration

change the hostname of a router:

=====

```
#conf t
config# hostname Mumbai
config# exit
```

=====

First Router:

```
Router>enable
Router# configure terminal
Router(config)# interface fastethernet0/0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no shutdown
enable interface
Router(config-if)#exit
from current mode
Router(config)# interface serial0/0/0
Router(config-if)#ip address 140.10.0.1 255.255.0.0
Router(config-if)#no shutdown
Router(config-if)#end
Router# show running-config
DRAM configuration
Router# show ip interface brief
interface summary in tabular format
```

goto pc0 / pc1

click on desktop --- command prompt-

gateway(should ping)	c:\>ping 10.0.0.1	ping to
	c:\>ping 10.0.0.3	
serial interface(should ping)	c:\>ping 140.10.0.1	ping to

Second Router:

```
Router>enable
Router# configure terminal
Router(config)# interface fastethernet0/0
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)# interface serial0/0/0
Router(config-if)#ip address 140.10.0.2 255.255.0.0
Router(config-if)#no shutdown
Router(config-if)#clock rate 64000                                #to set
the clock rate between two routers. In packet tracer
Router(config-if)#end or ctrl+z                                    clock
rate command is optional.
Router# show running-config                                         #shows
the DRAM configuration
Router# show ip interface brief
```

Basic Commands:

```
show startup-config          #shows saved configuration data(NVRAM)
write
copy running-config startup-config
show clock                   #shows router's clock
clock set 12:15:30 23 Sep 2025 #to set router's date & time
show version                 #shows hw, software , firmware details,
also it shows configuration register information(value)
```

8. show ip protocols	#shows routing protocol details
9. show protocols	#shows Ip configuration of interfaces
10. show history	#shows last 10 previously used commands
11. show terminal	#shows terminal setting details
12. show flash	# shows flash details(IOS filename, size
etc)	
13. del flash	#to delete Flash memory information
14. erase startup-config	#to delete saved configuration

15. description Admin to understand the use / purpose of the links

```
config# int fa0/0
description " this is LAN link for HR Dept"
```

=====

IP Routing:

The process of transferring packets from source to destination through inter-network it consists of two phases

1. Identify the available networks in the topology, and define the best path to reach to that destination.
2. Transferring the packets over the defined path.

Types of Routing:

Static Routing
Default Routing
Dynamic Routing

Static Routing:

- a. Routing is done manually by Administrator.
- b. Suitable for small internetworks
- c. More secure of routing.
- d. more burden will be on admin, less on Router's CPU & RAM

Static Routing can be done in 2 ways

- 1. Next Hop
- 2. Self Exit

IP Routing:

Routing: is a process of transferring packets from source to destination through an internetwork

It consist of two tasks:

- 1. searching for the available nws & defining best path.
- 2. forwarding the packets over the defined path.

Types of Routing

- 1. Static
- 2. Default
- 3. Dynamic

Types of Routing:

- 1. Static

to different nws

- 1. Administrator is responsible to define paths to reach
- 2. More workload on administrator
- 3. Less burden on Router's CPU and RAM (usage)
- 4. More secure
- 5. Administrator should have complete knowledge of

internetwork topology

- 6. Suitable for small internetworks
- 7. Change in any nw leads to redefine entries in all

routers for that nw.

Static routing can be done in 2 ways

- 1. Self exit method(directly connected)
- 2. Next Hop Method

Next Hop Method:

```
#conf t
config# ip route    <unknown nw>    <subnet mask>    <next hop ip>
```

Self exit method(directly connected)

```
#conf t
config# ip route    <unknown nw>    <subnet mask>    <Self Serial interface
name>
```

To remove Static Routing:

```
conf t
config# no ip route    <unknown nw>    <subnet mask>    <next hop ip>
```

Day2:

=====

Router: L3 device, comes with its IOS
never allow B'cast (by default), which helps reduce unnecessary network congestion.

- 1. Internetwork comm
- 2. Path Selection

3. Packet Switching

4. Packet Filtering

Routing

Router's Internal Components:

ROM Monitor)	POST, Boot loader, Mini IOS, ROMMON (ROM	
DRAM	Dynamic RAM	temporary memory
configuration (running-config)		contains running
NVRAM Non-Volatile RAM	Permanent memory	contains saved configuration
(startup-config)		
Flash	permanent memory	contains IOS of router
Circuit Board		
CPU		
configuration-register		

```

Router modes
Router>?          user execution mode          ssh telnet enable disable
ping exit
enable
Router#?          privilege mode              copy ping, show delete
configure terminal
Router(config)#?  Global configuration mode    hostname, to do global
configuration in the router
interface fastethernet0/0
3. Router(config-if)#? interface mode
ip address 10.0.0.1 255.0.0.0

Router(config-if)# no shutdown          to enable the
port

4. Router(config-line)# line mode
5. Router(config-router)# routing mode

```

```

int serial0/0/0
clock rate ?

```

```

exit          interface
exit          global
#show running-config
#show ip interface brief or #sh ip int bri

```

First Router:

```

=====
Basic Setup Mode: no
Router>enable
Router# configure terminal          #to enter into
global conf mode
Router(config)# interface fastethernet0/0          #to enter into
interface mode

```

```

Router(config-if)# ip address 10.0.0.1 255.0.0.0
Router(config-if)# no shutdown                                     #to enable to
interface                                                         #to exit from
Router(config-if)# exit
current mode
Router(config)# interface serial0/0/0
Router(config-if)# ip address 32.0.0.1 255.0.0.0
Router(config-if)# no shutdown
Router(config-if)# end                                           #to jump to
privilege mode
Router# show running-config                                       # to view running
configuration (DRAM)
Router# show ip interface brief                                   # to view
interface/ip status (tabular format)

```

Middle Router:

```

=====
Basic Setup Mode: no
Router>enable
Router# configure terminal
Router(config)# interface fastethernet0/0
Router(config-if)# ip address 172.16.0.1 255.255.0.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config-if)# ip address 32.0.0.2 255.0.0.0
Router(config-if)# no shutdown
Router(config)# interface serial0/0/1
Router(config-if)# ip address 34.0.0.1 255.0.0.0
Router(config-if)# no shutdown
Router(config-if)# clock rate 64000
Router(config-if)# end
Router# show running-config
Router# show ip interface brief

```

Last Router:

```

=====
Basic Setup Mode: no
Router>enable
Router# configure terminal
Router(config)# interface fastethernet0/0
Router(config-if)# ip address 192.168.0.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config-if)# ip address 34.0.0.2 255.0.0.0
Router(config-if)# no shutdown
Router(config-if)# end
Router# show running-config
Router# show ip interface brief

```

basic commands

```

#show running-config      or #show run    shows the running configuration (DRAM--
temp data)
2. #show ip interface brief  shows the ip and interface status in tabular format
3. #show clock              shows router's clock
4. #show terminal
5. #show history
6. #show startup-config     or  #show start  shows the saved configuration (NVRAM)
7. #copy running-config    startup-config  copying from DRAM to NVRAM
   #copy run start

```

```

8. #write
   #wr

```

```
To set hostname of router
#conf t
config# hostname Mumbai
```

```
#show run
#show start
```

9.reload reboot the router

10. show flash shows the router IOS details

11. erase startup-config to delete the saved configuration

12. show version shows hw, software firmware and configuration
register details

```
=====
=====
```

Routing: v imp
Routing is a process of transferring packets from source to destination through an internetwork.

It consists of two tasks

- a. Searching for the available networks and defining the best path
- b. forwarding packet on the defined paths.

Routing can be done in 3 ways

Static
Dynamic
Default

Static Routing:

- a. Self exit method/ directly connected (preferred method)
- b. Next hop method

1. administrator is responsible for searching nws and defining paths
2. good for small internetworks
3. Adding or removing network is time consuming process
4. More secure
5. more burden on administrator, less on router's CPU/RAM
6. Administrator should have complete knowledge of entire nw topology

Static Routing using Next hop method:

Mumbai:

```
#show ip route
```

```
#conf t
```

```
Format:      #ip route  unknown nw  subnet mask          next hop ip
config# ip route 172.16.0.0    255.255.0.0          32.0.0.2
config# ip route 192.168.0.0   255.255.255.0       32.0.0.2
config# ip route 34.0.0.0      255.0.0.0           32.0.0.2
```

```
# do show ip route          #do --> allowing you to
run privilege mode command from any mode
```

```
C 10.0.0.0/8
C 32.0.0.0/8
S 172.16.0.0
S 192.168.0.0
```

S 34.0.0.0

=====

Middle Router:

=====

>enable

Pune# show ip route

```
C      172.16.0.0
C      32.0.0.0
C      34.0.0.0
```

```
unknown      10.0.0.0
              192.168.0.0
```

#conf t

config# ip route 10.0.0.0 255.0.0.0 32.0.0.1

config# ip route 192.168.0.0 255.255.255.0 34.0.0.2

config# do show ip route

```
C      172.16.0.0
C      32.0.0.0
C      34.0.0.0
S      10.0.0.0
S      192.168.0.0
```

Last Router:

=====

#conf t

config# ip route 10.0.0.0 255.0.0.0 34.0.0.1

config# ip route 172.16.0.0 255.255.0.0 34.0.0.1

config# ip route 32.0.0.0 255.0.0.0 34.0.0.1

config# do show ip route

=====

Static Routing using Self Exit method:

Format: #ip route unknown nw subnet mask self exit

interface name

Mumbai Router:

=====

#show ip route

#conf t

config# ip route 172.16.0.0 255.255.0.0 s0/0/0

config# ip route 192.168.0.0 255.255.255.0 s0/0/0

config# ip route 34.0.0.0 255.0.0.0 s0/0/0

config# do show ip route

Pune Router:

=====

#conf t

config# ip route 10.0.0.0 255.0.0.0 s0/0/0

config# ip route 192.168.0.0 255.255.255.0 s0/0/1

config# do show ip route

BLR Router:

=====

#show ip route

#conf t

config# ip route 172.16.0.0 255.255.0.0 s0/1

config# ip route 10.0.0.0 255.0.0.0 s0/1

config# ip route 32.0.0.0 255.0.0.0 s0/1

config# do show ip route

goto Pcs, check the connectivity between PCs

=====

goto Pcs, check the connectivity between PCs